

```
import java.util.Scanner;
```

```
public class Calculator {
```

```
    public static double add(double a, double b) {  
        return a + b;  
    }
```

```
    public static double subtract(double a, double b) {  
        return a - b;  
    }
```

```
    public static double multiply(double a, double b) {  
        return a * b;  
    }
```

```
    public static double divide(double a, double b) {  
        if (b == 0) {  
            System.out.println("Error: Division by zero is not allowed.");  
            return Double.NaN;  
        }  
        return a / b;  
    }
```

```
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
char choice;
```

```
do {
```

```
    System.out.println("\n===== JAVA CONSOLE CALCULATOR =====");
```

```
    System.out.println("1. Addition");
```

```
    System.out.println("2. Subtraction");
```

```
    System.out.println("3. Multiplication");
```

```
    System.out.println("4. Division");
```

```
    System.out.print("Enter your choice (1-4): ");
```

```
    int option = sc.nextInt();
```

```
    System.out.print("Enter first number: ");
```

```
    double num1 = sc.nextDouble();
```

```
    System.out.print("Enter second number: ");
```

```
    double num2 = sc.nextDouble();
```

```
    switch (option) {
```

```
        case 1:
```

```
            System.out.println("Result = " + add(num1, num2));
```

```
            break;
```

```
        case 2:
```

```
            System.out.println("Result = " + subtract(num1, num2));
```

```
            break;
```

```
        case 3:
```

```
            System.out.println("Result = " + multiply(num1, num2));
```

```
break;
```

```
case 4:
```

```
double result = divide(num1, num2);
```

```
if (!Double.isNaN(result)) {
```

```
    System.out.println("Result = " + result);
```

```
}
```

```
break;
```

```
default:
```

```
    System.out.println("Invalid choice! Please select 1-4.");
```

```
}
```

```
System.out.print("\nDo you want to perform another calculation? (Y/N): ");
```

```
choice = sc.next().charAt(0);
```

```
} while (choice == 'Y' || choice == 'y');
```

```
System.out.println("\nThank you for using the Calculator!");
```

```
sc.close();
```

```
}
```

```
}
```